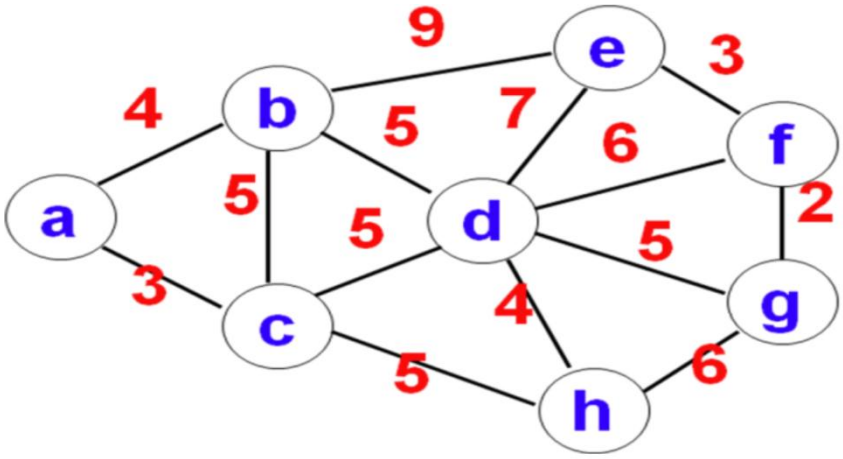
作业 14：图的最小生成树

请对下面的无向带权图

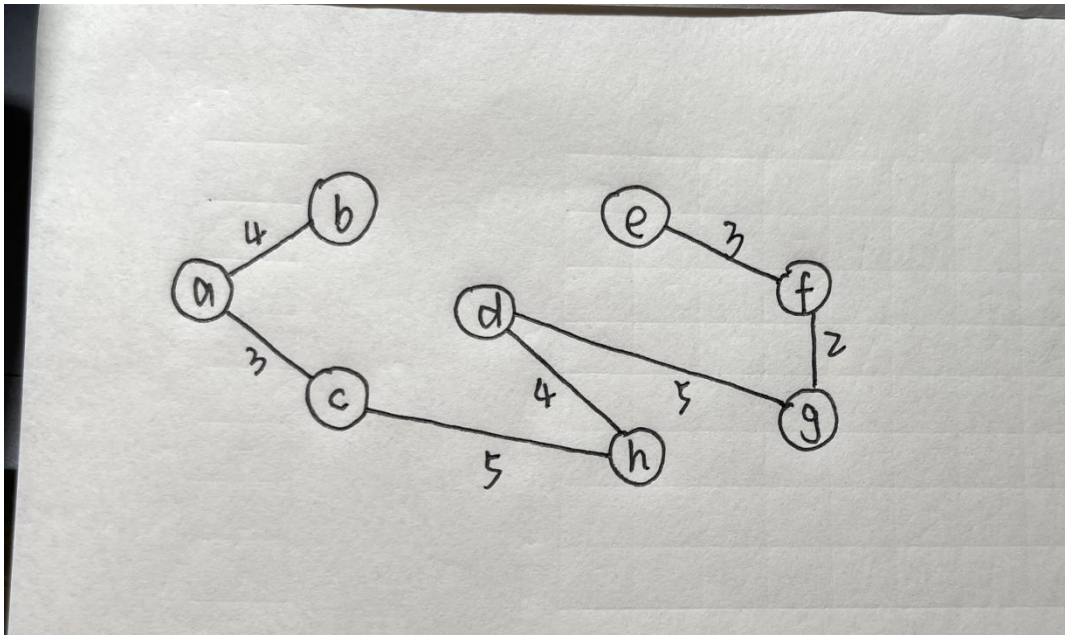


（1）写出邻接矩阵，并按普里姆算法求最小生成树。

邻接矩阵

0	a
1	b
2	c
3	d
4	e
5	f
6	g
7	h

	a	b	c	d	e	f	g	h
a	0	4	3	0	0	0	0	0
b	4	0	5	5	9	0	0	0
c	3	5	0	5	0	0	0	5
d	0	5	5	0	7	6	5	4
e	0	9	0	7	0	3	0	0
f	0	0	0	6	3	0	2	0
g	0	0	0	5	0	2	0	6
h	0	0	5	4	0	0	6	0



普里姆算法：

①

a	b	c	d	e	f	g	h
-1	-1	-1	-1	-1	-1	-1	-1
p[0]	p[1]	p[2]	p[3]	p[4]	p[5]	p[6]	p[7]
0	∞	∞	∞	∞	∞	∞	∞
d[0]	d[1]	d[2]	d[3]	d[4]	d[5]	d[6]	d[7]

S={ }　Q={a, b, c, d, e, f, g, h}.

②

a	b	c	d	e	f	g	h
-1	0	0	-1	-1	-1	-1	-1
p[0]	p[1]	p[2]	p[3]	p[4]	p[5]	p[6]	p[7]
0	4	3	∞	∞	∞	∞	∞
d[0]	d[1]	d[2]	d[3]	d[4]	d[5]	d[6]	d[7]

S={a}　Q={b, c, d, e, f, g, h}.

③

a	b	c	d	e	f	g	h
-1	0	0	2	-1	-1	-1	2
p[0]	p[1]	p[2]	p[3]	p[4]	p[5]	p[6]	p[7]
0	4	3	5	∞	∞	∞	5
d[0]	d[1]	d[2]	d[3]	d[4]	d[5]	d[6]	d[7]

S={a, c}　Q={b, d, e, f, g, h}.

④

a	b	c	d	e	f	g	h
-1	0	0	2	1	-1	-1	2
p[0]	p[1]	p[2]	p[3]	p[4]	p[5]	p[6]	p[7]
0	4	3	5	9	∞	∞	5

S={a, c, b}　Q={d, e, f, g, h}.

⑤

a	b	c	d	e	f	g	h
-1	0	0	7	1	-1	7	2
p[0]	p[1]	p[2]	p[3]	p[4]	p[5]	p[6]	p[7]
0	4	3	4	9	∞	6	5
d[0]	d[1]	d[2]	d[3]	d[4]	d[5]	d[6]	d[7]

S={a, c, b, h}　Q={d, e, f, g}.

⑥

a	b	c	d	e	f	g	h
-1	0	0	7	3	3	3	2
p[0]	p[1]	p[2]	p[3]	p[4]	p[5]	p[6]	p[7]
0	4	3	4	7	6	5	5
d[0]	d[1]	d[2]	d[3]	d[4]	d[5]	d[6]	d[7]

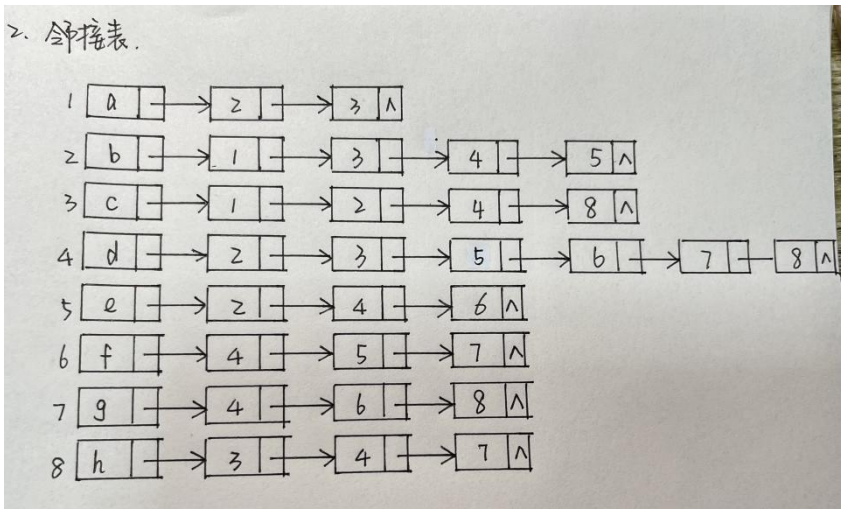
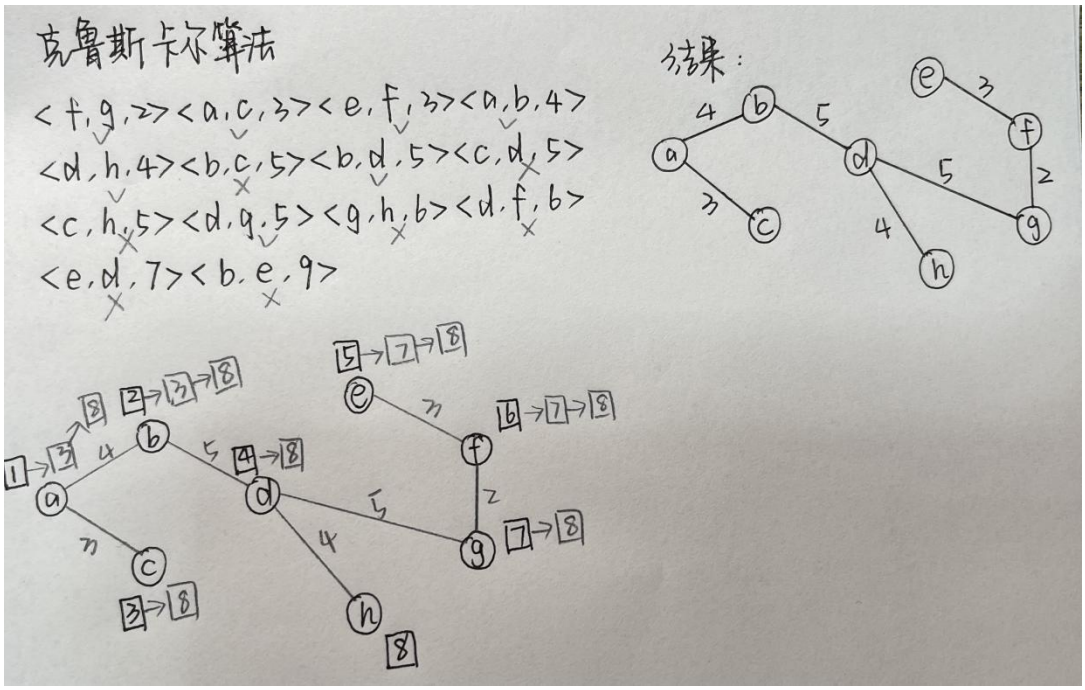
S={a, c, b, h, d}　Q={e, f, g}

⑦

a	b	c	d	e	f	g	h
-1	0	0	7	5	6	3	2
p[0]	p[1]	p[2]	p[3]	p[4]	p[5]	p[6]	p[7]
0	4	3	4	3	2	5	5
d[0]	d[1]	d[2]	d[3]	d[4]	d[5]	d[6]	d[7]

S={a, c, b, h, d, g}　Q={e, f}.

⑧ S={a, c, b, h, d, g, f}　Q={e}

⑨ S={a, c, b, h, d, g, f, e}　Q={ }

（2）写出邻接表，并按克鲁斯卡尔算法求其最小生成树。